

JAVA BACKEND ENGINEERING / GEOSPATIAL SYSTEMS

Alireza Nobakht

Java Backend Developer | Spring Boot Engineer

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PROFESSIONAL PROFILE

Java Backend Developer with **2+ years of software engineering experience** building Spring Boot APIs, Hibernate/JPA data services, geospatial platforms, and Android field applications. Experienced in Java, Spring Boot, RESTful services, PostgreSQL/PostGIS, clean architecture, database optimization, offline synchronization, and connected-device workflows. Combines backend engineering with practical GPS/IMU and embedded-systems experience to deliver reliable software for field operations.

01 TECHNICAL SKILLS

JAVA & SPRING

Java, Spring Boot, Spring Framework, Spring MVC, REST APIs, dependency injection, OOP, async and concurrent programming

PERSISTENCE & DATA

JPA / Hibernate ORM, PostgreSQL, PostGIS, SQL Server, database schemas, spatial indexing, query optimization, Redis caching

ARCHITECTURE & APIS

Clean Architecture, DDD, SOLID, design patterns, layered applications, DTO mapping, validation, standardized errors, OpenAPI documentation

MOBILE & WEB

Java Android SDK, Kotlin / Android, Thymeleaf, React, JavaScript, HTML, CSS, Leaflet, OpenLayers, offline-first sync

DELIVERY & QUALITY

Git, Docker, GitHub Actions, CI/CD pipelines, Linux fundamentals, test planning, monitoring, production debugging, code review

CONNECTED ENGINEERING

STM32, C/C++, Qt, C#, UART, SPI, I2C, GPS / IMU, serial protocols, multithreading, real-time processing, encrypted transfer

02 PROFESSIONAL EXPERIENCE

Software Developer & Embedded Systems Engineer

Feb 2024 - Present

Hiromap (Hirongar Pars Company) | hiromap.com | Tehran, Iran

- **Developed a Java/Spring Boot geospatial platform** with PostgreSQL/PostGIS for storing, processing, and querying survey coordinates, maps, and field measurements.
- **Designed RESTful APIs** for data ingestion, retrieval, project workflows, and map rendering, using Hibernate ORM and spatial indexing for scalable dataset access.
- **Integrated React, Leaflet, and OpenLayers** to deliver interactive map visualization, survey-point management, and topographic-layer overlays.
- **Built a Java Android field-survey application** with GPS tracking, offline caching, validation, and synchronization to a Spring Boot server in limited connectivity environments.
- **Created a Spring Boot + Thymeleaf admin dashboard** for survey projects, users, and field reports; real-time validation and automated coordinate correction reduced manual-entry errors by **40%**.
- **Produced C++/Qt and C# device-diagnostics tools** for 115200-baud serial streams with zero observed data loss and 40% lower memory use, supporting GPS/IMU workflows.

Technical Instructor & Educational Consultant

Aug 2022 - Jun 2023

Ghalamchi Educational Institute | Tehran, Iran

- **Mentored students during a 10-month engagement** in programming fundamentals, algorithms, problem solving, and software development practices.
- **Prepared structured lessons, exercises, and debugging walkthroughs** that strengthened technical communication, collaboration, and systematic problem solving.

SELECTED JAVA ENGINEERING WORK

Projects, Education & Development

Spring Boot APIs, geospatial data services, Android field software, and connected engineering.

JAVA BACKEND DEVELOPER
SPRING BOOT ENGINEER

Geospatial Data Management Platform

Apr 2025

Survey-data management and visualization

Java · Spring Boot · Hibernate ORM · PostgreSQL / PostGIS · REST · React · Leaflet

- **Built a Spring Boot + PostgreSQL/PostGIS platform** to store, process, and visualize survey points, coordinates, field measurements, and topographic data.
- **Implemented REST services** for data ingestion, retrieval, processing, and map rendering, backed by Hibernate ORM and optimized spatial queries.
- **Connected React with Leaflet/OpenLayers** for interactive map layers, survey-point management, and efficient access to large geospatial datasets.

Field Survey Data Collection App

Jul 2024

Offline-capable mobile surveying workflow

Java / Android SDK · Spring Boot · REST APIs · GPS · Offline Storage · Thymeleaf

- **Developed a Java Android field application** for recording geotagged coordinates, notes, images, and field observations directly on site.
- **Implemented GPS tracking, local caching, and secure REST synchronization** with a central Spring Boot service to support limited connectivity environments.
- **Added validation and coordinate-correction workflows**, reducing manual input errors by **40%** and improving field-data reliability.

Device Diagnostics & Cross-Platform Transfer

2024 - 2025

Real-time tooling and device integration

- Built C++/Qt and C# applications for streaming, filtering, visualizing, recording, and analyzing serial data.
- Supported CSV, JSON, and binary exports with asynchronous buffering and configurable communication parameters.
- Engineered encrypted Bluetooth/Wi-Fi file transfer with progress, resume, retry, and chunking for files larger than 100 MB.

115200	99.5%	35%
baud processing	transfer success	faster transfer

Qt · C# · Kotlin / Android · Encryption

Surveying Rover Firmware

Jul 2024

Navigation and real-time data collection

- Led STM32 firmware development for a 100 Hz control loop with PID motor control, GPS/IMU fusion, Kalman filtering, and power management.
- Established UART and SPI telemetry with fault recovery and error correction for difficult RF conditions.
- Achieved +/-2 cm accuracy, 99.9% transmission reliability over 500 m, and 8+ hours of runtime.

STM32 · C/C++ · GPS / IMU · PID · Kalman Filter

EDUCATION

B.Sc. Computer Engineering

K.N. Toosi University of Technology
Sep 2022 - Present

Expected graduation: 2026. Teaching Assistant for core programming courses. Study areas include algorithms, databases, software engineering, operating systems, networks, embedded systems, and OOP.

PROFESSIONAL FOCUS

Backend & Geospatial Systems

Project-based development with Spring Boot, Hibernate ORM, REST APIs, PostgreSQL/PostGIS, Android, and clean architecture.

Ongoing study: scalable backend design, database performance, software architecture, cloud fundamentals, and maintainable API design.

LANGUAGES & PRACTICES

Persian - Native

English - Intermediate / Professional

Technical documentation, independent work, cross-functional collaboration, code review, production debugging, systematic problem solving, and clear stakeholder communication.

BACKEND APIS

Spring Boot REST services, validation, standardized responses, and role-aware workflows.

SPATIAL DATA

PostgreSQL/PostGIS models, survey points, spatial indexing, and map-layer access.

FIELD MOBILITY

Java Android GPS capture, offline storage, synchronization, and data correction.

DEVICE WORKFLOW

UART/SPI telemetry, diagnostics, serial processing, and encrypted transfer tooling.